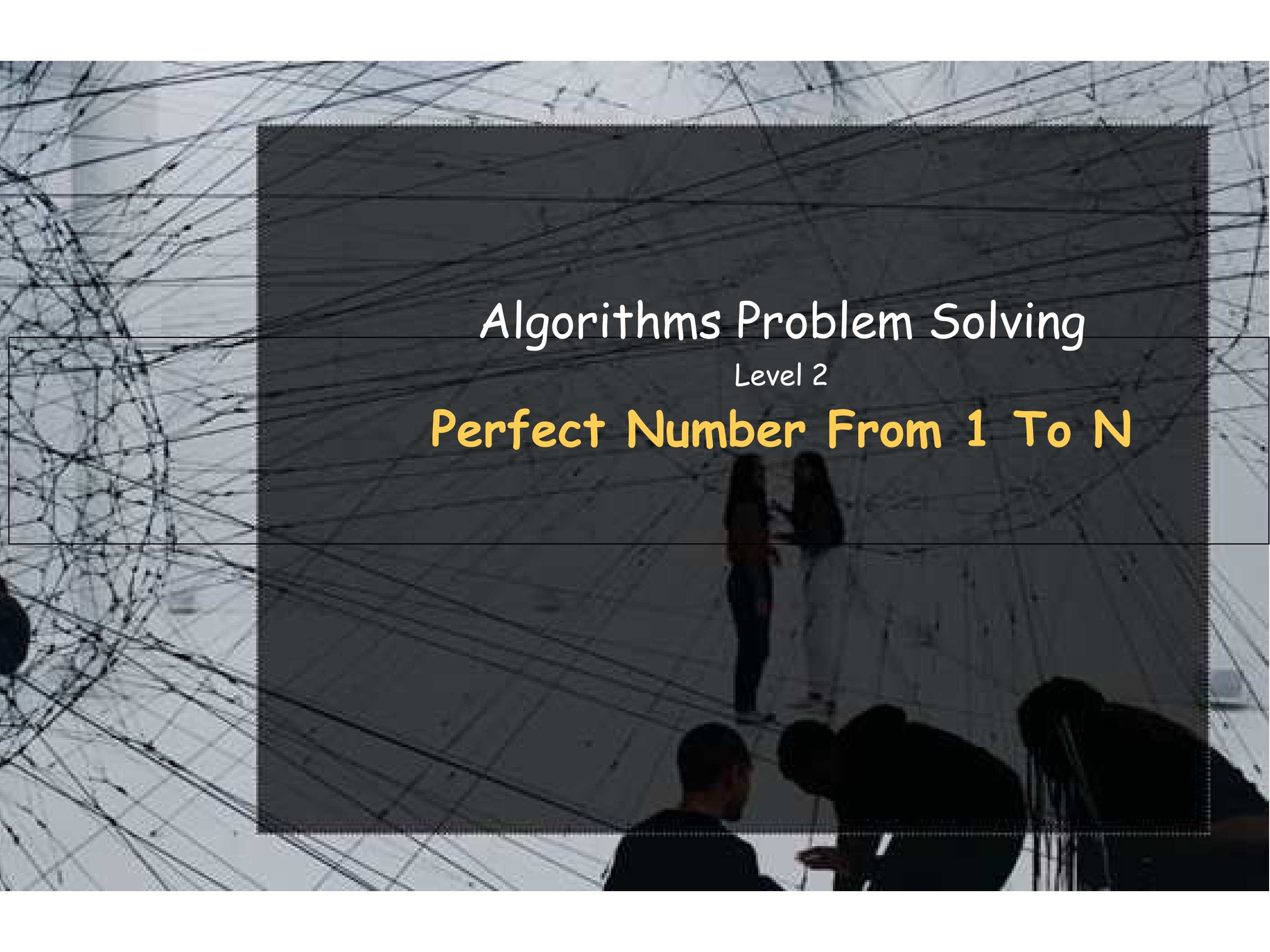


Algorithms Problem Solving

Level 2

Perfect Number From 1 To N



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Definition

- ◆ A Perfect Number is a positive number that is equal to the sum of all its proper divisors, excluding the number itself.
- ◆ This program reads a number N and prints all Perfect Numbers from 1 to N .

Explanation

- ◆ ``ReadNumber()`` reads a positive number from the user using a ``do-while`` loop.
- ◆ ``CheckPerfect()`` checks whether a number is Perfect by calculating the sum of its proper divisors.
- ◆ The ``for`` loop inside ``CheckPerfect()`` checks all numbers from ``1`` to ``Numb - 1``.
- ◆ If ``Numb % i == 0``, then ``i`` is a divisor and is added to ``Sum``.
- ◆ ``Numb == Sum`` returns ``true`` when the number is Perfect.
- ◆ ``PrintResult()`` loops through every number from ``1`` to `N`.
- ◆ Each number is passed to ``CheckPerfect()``.
- ◆ If the result is ``true``, the number is printed.

Steps in main()

1. Read a positive number N.
2. Start a loop from 1 to N.
3. Check each number using `CheckPerfect()`.
4. Find all proper divisors of the current number.
5. Calculate their sum.
6. Compare the sum with the current number.
7. Print the number if it is Perfect.

💡 Example

◆ Input:

`500`

◆ The program checks every number from `1` to `500`.

◆ The Perfect Numbers found are:

`6`

`28`

`496`

➡ Result

`6`

`28`

`496`

Note

- ◆ The program uses the same `CheckPerfect()` function from the previous lesson.
- ◆ The difference is that instead of checking only one number, `PrintResult()` checks every number from `1` to `N`.

⊘ Warning

- ◆ The program checks every number from `1` to `N`, and for each number it checks all smaller numbers as possible divisors.
- ◆ This means there are nested loops, so the program can become slow for very large values of N.



Comparison

▨ Previous Program

- ✓ Checks one specific number.
- ✓ Example: `28 → Perfect`.

▨ This Program

- ✓ Checks all numbers from `1` to N.
- ✓ Example: `500 → 6, 28, 496`.

✦✦ Summary

- ✓ Read a positive number N.
- ✓ Check every number from 1 to N.
- ✓ Use `CheckPerfect()` to test each number.
- ✓ Add all proper divisors of each number.
- ✓ Print the number when the divisor sum equals the number.
- ✓ For `N = 500`, the Perfect Numbers are `6`, `28`, and `496`.

Thank you 😊